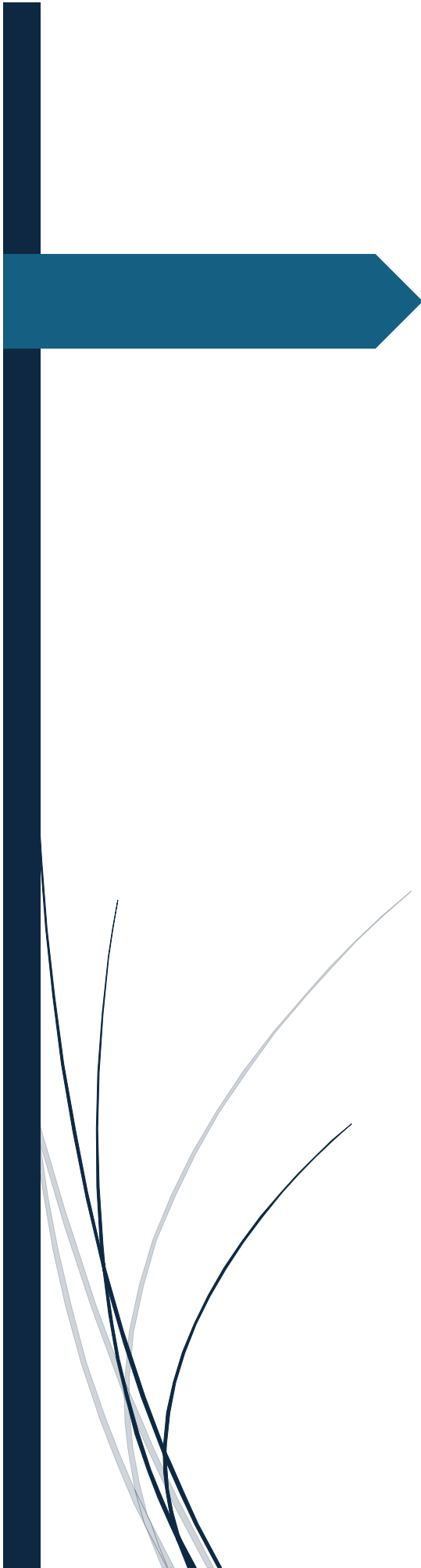




Assignment 2

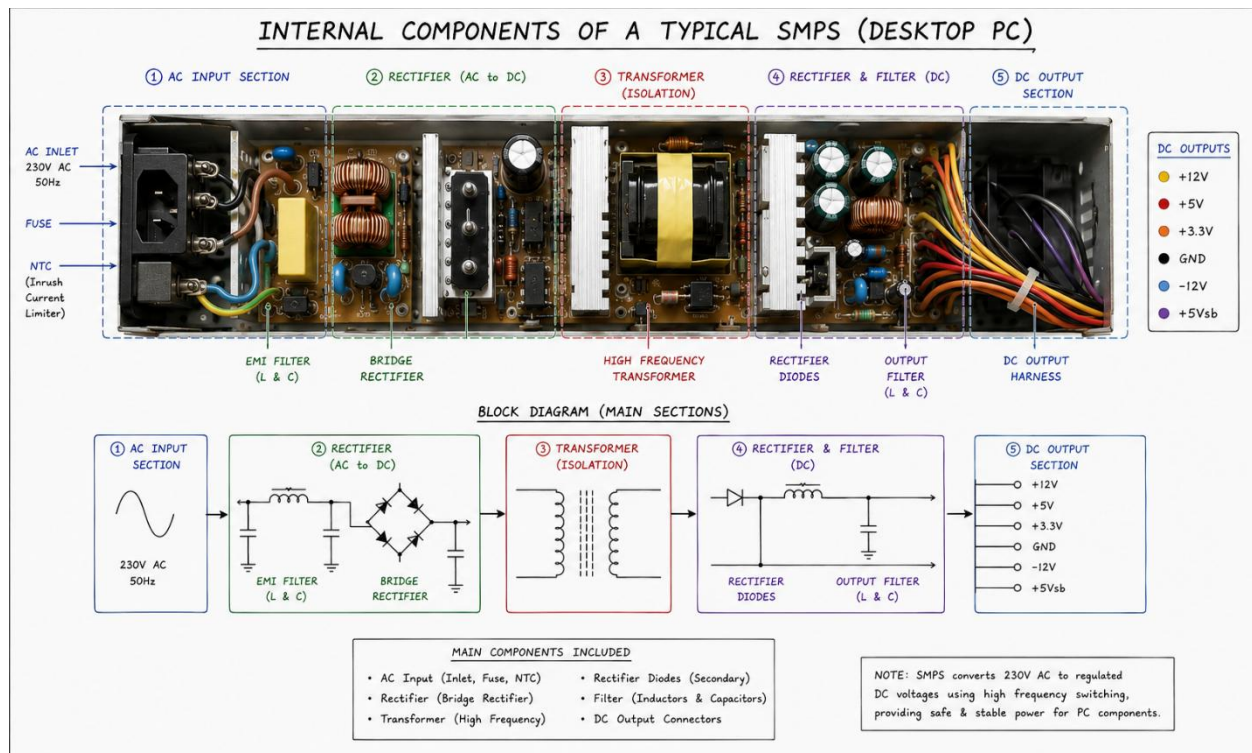
SMPS & UPS, Cables & Connector

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

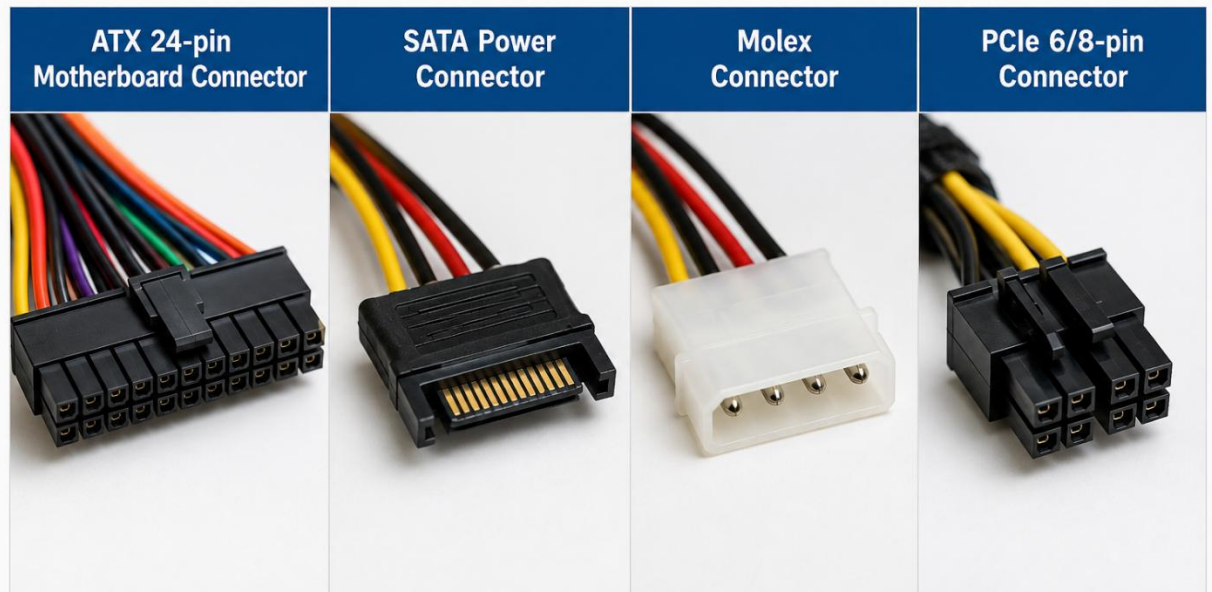
Device	Power Supply	Why does it need it?
➤ Desktop Computer	SMPS	It gives the computer the right amount of power to work safely.
➤ Printer	SMPS	It supplies the power needed for the printer to work properly.
➤ Wi-Fi Router	SMPS	It changes the electricity into the lower power needed by the router.
➤ Desktop Computer	UPS	It gives backup power when the electricity goes off.
➤ Wi-Fi Router	UPS	It keeps the router working for some time during a power cut.

➤ **Simple difference: SMPS gives the device the right power, while UPS gives backup power when the electricity goes out.**

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



1. ATX 24-pin Connector

- **Devices it powers:** Motherboard
- **Safety tip:** Make sure the connector is fitted properly and the clip locks in place. Do not force it.

2. SATA Power Connector

- **Devices it powers:** HDD, SSD, and DVD drive
- **Safety tip:** Connect it gently in the correct direction. Do not bend or force the connector.

3. Molex Connector

- **Devices it powers:** Older hard drives, DVD drives, and some fans
- **Safety tip:** Check the direction before connecting and do not force it.

4. PCIe 6/8-pin Connector

- **Devices it powers:** Graphics card (GPU)
- **Safety tip:** Make sure it is fully connected and locked before turning on the computer.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

❖ If I were setting up a new gaming PC, I would connect the SMPS in these steps:

1. First, turn off the PC and unplug the SMPS from the power socket.

2. Connect the motherboard: I would take the **24-pin ATX cable** from the SMPS and plug it into the main power connector on the motherboard.

3. Connect the CPU: Next, I would use the **4+4-pin CPU cable** and connect it near the CPU socket on the motherboard.

4. Connect the graphics card: I would use the **PCIe 6/8-pin cable** and plug it into the graphics card to give it power.

5. Connect the storage: For a **SATA HDD or SSD**, I would use the **SATA power cable** from the SMPS and connect it to the drive.

6. Check all the cables: I would make sure all the connectors are properly plugged in and not loose.

7. Turn on the PC: Finally, I would connect the SMPS to the power socket and turn on the computer to check that everything works properly.